MIPS

FPGA Module
User's Guide, Rev 1.0
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Introduction

The MIPS FPGA module is an internal PCB that can be installed in MIPS systems to enable advanced capabilities. The module is inserted into the internal MIPS bus system between modules and the MIPS controller, enabling a number of different hardware-accelerated functions.

The MIPS FPGA module contains:

Eight channels of ADC input, ±10 V, 14-bit resolution
Eight high-speed digital I/O lines
Two-channel PLL frequency generator
On-board configuration memory

This FPGA module uses a Xilinx Spartan 7 FPGA and is programmed using the Xilinx Vivado development environment.

The following section describes different FPGA module configurations. Refer to your MIPS order or the About Your System documentation supplied with your MIPS system for your specific configuration.

ADC to DC Bias Reflector

This operating mode reads voltages from the eight ADC inputs and uses those analog values as setpoints to control eight DC bias channel outputs. The entire process is performed in hardware by the FPGA and requires no MIPS CPU resources.

The eight input channels are scanned, their values are converted into a valid range for the DC bias module, and the results are sent to the DACs on the DC bias module. ADC values can be read by MIPS at any time, and the reflection function can be enabled or disabled.

Individual channels can be masked, which removes them from the FPGA-controlled update cycle and allows those channels to be controlled independently through the MIPS interface.

Reflection Process

The reflection process executes the following steps on each cycle:

1. Scans the ADC registers and populates an internal 16-bit ADC array with the raw ADC data.
2. Calculates the required DAC values from the ADC array using the M (slope) and B (offset) calibration arrays:

$DAC[i] = (ADC[i] \times M[i]) / 1024 + B[i]$

where i is the channel index (1 through 8).

3. Writes the DAC array values to the DC bias module configured for reflection.

Note: Your system is shipped with a fully functional control panel UI that allows operation of the reflection mode without the need for any of the commands listed below. These commands are included here for completeness. Please contact GAA Custom Electronics before making any changes.

Commands

All commands are case-sensitive. The following commands are available for configuring and controlling the ADC to DC Bias Reflector operating mode.

SFPGACFG,*value*
Sets the FPGA configuration word. *Value* is a hex integer; the individual bit definitions are listed in Table 1 below.

GFPGACFG
Returns the current FPGA configuration word in hex format.

SFPGABC,*value*
Enables or disables FPGA bus control. *Value* = TRUE to enable, FALSE to disable.

GFPGABC
Returns the FPGA bus control status, TRUE or FALSE.

SFPGAP,*value*
Enables or disables the ADC-to-DAC reflection process. *Value* = TRUE to start the process (the FPGA reads ADC registers and calculates DAC values); FALSE to stop it.

GFPGAP
Returns the current status of the FPGA reflection process, TRUE or FALSE.

SFPGASCN,*value*
Enables or disables the ADC scanning process. *Value* = TRUE to enable, FALSE to disable.

GFPGASCN
Returns the ADC scanning process status, TRUE or FALSE.

SFPGAADD,*value*
Sets the SPI address used to communicate with the DC bias DAC registers. *Value* = the desired address as an integer.

GFPGAADD
Returns the current SPI address value.

SFPGAARY,*Array*,*Index*,*Value*
Writes a value to one of the FPGA's internal 16-bit data registers used by the ADC-to-DC Bias reflector. *Array* selects the target array (M, B, ADC, or DAC). *Index* is the array element (1 through 8). *Value* is the 16-bit integer to write.

GFPGAARY,*Array*,*Index*
Reads a value from one of the FPGA's internal 16-bit data registers. *Array* selects the source array (M, B, ADC, or DAC). *Index* is the array element (1 through 8).

SADCMSK,*value*
Sets the ADC channel scan mask. The ADC scanning process can read all 8 channels or a subset; this mask register selects which channels are active. *Value* is a hex bitmask where bits 0 through 7 correspond to channels 1 through 8. A set bit masks (disables) the corresponding channel. For example, a value of 21 (hex) masks channels 1 and 6.

GADCMSK
Returns the current ADC channel scan mask value in hex format.

SDACMSK,*value*
Sets the DAC channel update mask. The DAC update process can write all 8 channels on the DC bias module or a subset; this mask selects which channels are updated. *Value* is a hex bitmask where bits 0 through 7 correspond to channels 1 through 8. A set bit masks (disables) the corresponding channel. For example, a value of 21 (hex) masks channels 1 and 6.

GDACMSK
Returns the current DAC channel update mask value in hex format.

RADCALL
Reads and reports all 8 ADC channels. Values are returned in volts.

GADCCH,*value*
Reads and returns the voltage of a single ADC channel. *Value* selects the channel (1 through 8).

SAVEFPGA
Saves all current FPGA settings to the FPGA module's non-volatile memory. The saved settings are automatically restored at startup.

RSTRFPGA
Restores previously saved settings from the FPGA module's non-volatile memory.

FPGACAL,*value*
Calibrates the ADC raw reading to voltage. This is a factory configuration command used during initial system setup.

FPGACALR,*value*
Performs the reflection mode calibration. This command reads the ADC calibration data and the DC bias module calibration data, then calculates all required internal parameters. *Value*

defines the first DC bias channel used and must be 1 or 9 depending on the hardware configuration. This is a factory calibration command.

Table 1, SFPGACFG Configuration Word Bit Definitions

Bit(s)	Function
0	FPGA control mode. 1 = FPGA module controls the bus.
1	Run mode. 1 = FPGA runs the reflection process.
2	ADC updates. 1 = FPGA updates the ADC register values in FPGA memory.
8	LDAC state when in FPGA control mode.
9	Board select for the controlled module.
10–12	SPI address of the controlled module.

Table 2, ADC to DC Bias Reflector Command Summary

Command	Parameters	Description
SFPGACFG	<value>	Sets the FPGA configuration word (hex). See Table 1 for bit definitions.
GFPGACFG		Returns the FPGA configuration word in hex format.
SFPGABC	<TRUE|FALSE>	Enables (TRUE) or disables (FALSE) FPGA bus control.
GFPGABC		Returns the bus control status, TRUE or FALSE.
SFPGAP	<TRUE|FALSE>	Enables (TRUE) or disables (FALSE) the ADC-to-DAC reflection process.
GFPGAP		Returns the reflection process status, TRUE or FALSE.
SFPGASCN	<TRUE|FALSE>	Enables (TRUE) or disables (FALSE) the ADC scanning process.
GFPGASCN		Returns the ADC scanning process status, TRUE or FALSE.
SFPGAADD	<value>	Sets the SPI address for the DC bias DAC registers (integer).
GFPGAADD		Returns the current SPI address value.
SFPGAARY	<Array>,<Index>,<Value>	Writes a 16-bit value to an internal FPGA register. Array = M, B, ADC, or DAC. Index = 1–8.
GFPGAARY	<Array>,<Index>	Reads a 16-bit value from an internal FPGA register. Array = M, B, ADC, or DAC. Index = 1–8.
SADCMSK	<value>	Sets the ADC channel scan mask (hex). Bits 0–7 correspond to channels 1–8; a set bit disables the channel.
GADCMSK		Returns the ADC channel scan mask in hex format.
SDACMSK	<value>	Sets the DAC channel update mask (hex). Bits 0–7 correspond to channels 1–8; a set bit disables the channel.
GDACMSK		Returns the DAC channel update mask in hex format.
RADCALL		Returns all 8 ADC channel voltages in volts.
GADCCH	<value>	Returns the voltage of a single ADC channel. Value = channel number (1–8).
SAVEFPGA		Saves all FPGA settings to non-volatile memory. Settings are restored automatically at startup.
RSTRFPGA		Restores saved settings from non-volatile memory.
FPGACAL	<value>	Factory command. Calibrates ADC raw reading to voltage.
FPGACALR	<value>	Factory command. Runs reflection mode calibration. Value = first DC bias channel (1 or 9 depending on hardware).